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Sensor system, assembly, method and computer program product for detecting events in an automatic dispensing process of discrete medicaments

Abstract

The invention relates to a sensor system, an assembly, a method and a computer program product for detecting events in an automatic dispensing process for dispensing discrete medicaments, wherein the sensor system comprises a light emitter, a light receiver and a control unit, wherein the light emitter forms a detection pair with the light receiver, wherein the control unit is configured for receiving the light input signals from the light receiver, wherein the control unit is further configured for operating in a dynamic mode in which the light output of the light emitter is adjusted in response to the light input values.

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Background/Summary

BACKGROUND

[0001] The invention relates to a sensor system, an assembly, a method and a computer program product for detecting events in an automatic dispensing process of discrete medicaments.

[0002] U.S. Pat. No. 10,173,830 B1 discloses a system which accommodates a series of medication dispensing containers, also known as ‘feeder units’ or ‘canisters’, for selectively dispensing an amount of said solid medications from one or more medication dispensing containers and for packing said dispensed amount. Each medication dispensing container holds an amount of solid substances specific to that respective medication dispensing container. Hence, together, the medication dispensing containers can dispense a wide variety of solid substances.

[0003] The medication dispensing container comprises an arranging body with a plurality of passageways, wherein each passageway is precisely dimensioned for accommodating said solid medications in a single row.

[0004] In use, the medication dispensing containers are docked to a holding device. When the medication dispensing container is correctly docked to the holding device, positioning columns of said holding device reach through detector openings into the medication dispensing container. Light sources and light detectors associated with said holding device align with the medication dispensing container and form detection pairs to detect the passage of medicaments through the passageways and through the outlet.

SUMMARY

[0005] A disadvantage of the known system is that the medication dispensing container and/or the holding device become polluted over time with dust or particles. The pollution can interfere with the transmission of light beams from the light sources towards the respective light detectors. If the medication dispensing container and the holding device are not cleaned in due time, the detection pairs may generate false detection result or will not be able to detect anything at all.

[0006] Additionally, the characteristics of the light sources and the light detectors may deviate slightly from their respective factory specifications or the tolerances in the factory specifications are not small enough for the intended application. This makes it difficult for the detection pairs to work together properly.

[0007] Moreover, the light source may be subject to light output degradation due to aging. At some point, the light output may be insufficient to properly detect medicaments passing the detection pair.

[0008] Finally, when pollution or damage, for example scratches, prevents a proper detection of medicaments, the pollution needs to be removed. The medication dispensing containers can be removed relatively easily for cleaning or repairs and cause minimal downtime. In contrast, the holding devices may be considerably more difficult to remove for cleaning. However, it is unknown to the pharmacist which one of the medication dispensing container and the holding device is polluted, or if both are polluted.

[0009] It is an object of the present invention to provide a sensor system, an assembly, a method and a computer program product for detecting events in an automatic dispensing process for dispensing discrete medicaments, wherein the reliability of the detection can be improved and/or

pollution can be removed more effectively.

[0010] According to a first aspect, the invention provides a sensor system for detecting events in an automatic dispensing process for dispensing discrete medicaments, wherein the sensor system comprises a light emitter for generating light output, a light receiver for converting light input into one or more light input signals indicative of light input values and a control unit that is operationally connected to said light emitter and said light receiver, wherein the light emitter forms a detection pair with the light receiver, wherein the control unit is configured for receiving the one or more light input signals from the light receiver, wherein the control unit is further configured for operating in a dynamic mode in which the light output of the light emitter is adjusted in response to the light input values.

[0011] The sensor system can be used in an assembly for detecting dispensing events of a feeder unit that is being docked to a feeder dock, as will be described in more detail later in this summary.

[0012] In the context of the present invention, the 'detection pair' is not limited to a combination of a single light emitter and a single light receiver. Instead, the light emitter may be part of or comprise an array of a plurality of light emitters, for example to form a light curtain. Similarly, the light receiver may be part of or comprise an array of a plurality of light receivers. As such, the array of light emitters and/or light receivers may be deemed the light emitter and light receiver of the detection pair.

[0013] By adjusting the light output, any negative effects of pollution between the light emitter and the light receiver of the detection pair, age degradation of the light emitter or slight variations in the characteristics of the light emitter and the light receiver of the detection pair can be at least partially compensated. In particular, the light output may be controlled such that the light input received by the respective light receiver is or remains within an optimal operating range or window for said light receiver. Hence, the detection pair can continue to detect events of in the automatic dispensing process reliably.

[0014] The control unit, in the dynamic mode, is configured for increasing the light output of the light emitter when the light input values decrease over time. Hence, even when the light beam between the light emitter and the light receiver of the detection pair is hindered by pollution, the light receiver can still receive the increased light output in order to reliably detect the events. Moreover, the light output of the light emitter may be used as an indicator of how polluted the sensor system, the feeder unit and/or the feeder dock are; e.g. the higher the light output, the higher the pollution. The increase in light output may be compared to a fixed value or a previously determined reference value, for example a light output level that has been determined during a prior calibration of the sensor system.

[0015] In one embodiment the control unit, in the dynamic mode, is configured for increasing the light output of the light emitter when the light input values drop below an input threshold. The input threshold may be chosen at a level which is within the optimal operating range or window of the light receiver, i.e. the range in which the light receiver is most sensitive to light.

[0016] Preferably, the input threshold is a predetermined input threshold value stored in a memory. The input threshold value may be chosen by the pharmacist or another operator, or it may be based on factory specifications of the light receiver.

[0017] The control unit is configured for switching between the dynamic mode and a calibration mode, wherein the input threshold is a calibration value that is determined based on one or more of the light input values during the calibration mode. Hence, the input threshold may be adjusted to best suit the conditions as measured during the calibration mode, for example taking into account variations in the light output, such as age degradation, and/or any pollution between the light emitter and the light receiver of the detection pair.

[0018] In a further embodiment the input threshold is equal to or less than sixty percent of a maximum light input signal of the light receiver pair. By choosing the input threshold below sixty percent, it can be ensured that the light receiver is properly triggered and/or activated by the light

input that is received.

[0019] In another embodiment the control unit, in the dynamic mode, is configured for storing the light input values continuously or at a regular interval. By storing the light input values continuously, any decrease in the light input be early and/or accurately. 10 values can detected Alternatively, storing the light input values at regular intervals may reduce the data consumed by storing the light input values. This can be, for example, at regularly scheduled times, after a certain period of use or other intervals.

[0020] In another embodiment the control unit, in the dynamic mode, is configured for increasing the light output of the light emitter up to an output threshold. Preferably, the output threshold is equal to or less than ninety percent of a maximum output of the light emitter. When said output threshold has been reached, it can be assumed that the sensor system, the feeder unit and/or the feeder dock have become too polluted to continue dispensing and detecting in a reliable manner.

[0021] According to a second aspect, the invention provides an assembly of the sensor system according to any one of the previously discussed embodiments and a feeder dock for docking a feeder unit, wherein the light emitter and/or the light receiver are provided in, on or at the feeder dock.

[0022] The assembly according to the second aspect of the invention includes the sensor system according to the first **30** aspect of the invention and thus has the same technical advantages, which will not be repeated hereafter.

[0023] In one embodiment the feeder unit comprises a dispensing mechanism for dispensing the discrete medicaments one-by-one, wherein the feeder dock comprises a dispensing drive that is arranged for engaging with the dispensing mechanism when said feeder unit is received at the feeder dock, wherein the control unit is operationally connected to the dispensing drive to operate the dispensing mechanism at dispensing intervals, wherein the control unit, in the dynamic mode, is configured for storing the light input values for the one or more light input signals that are received between the dispensing intervals. Hence, the light input values which are stored correspond to a light beam between the light emitter and the light receiver of the detection pair which was uninterrupted, or which at least was not interrupted or blocked by a discrete medicament at the time.

[0024] In another embodiment the feeder unit has an at least partially transparent housing, wherein the light emitter and the light receiver are positioned such that, when the feeder unit is received in the feeder dock, a light beam travelling between the light emitter and the light receiver travels through a transparent section of the at least partially transparent housing at least once. The transparent section can become polluted by dust or particles of the discrete medicaments that pass through and out of the feeder unit. By positioning the detection pair in such a way that the light beam passes through the transparent section, allows for the sensor system to not only detect the discrete medicaments, but also get an indication of pollution of the feeder unit.

[0025] In another embodiment the control unit is configured for switching between the dynamic mode and a calibration mode, wherein the control unit, in the calibration mode, is configured for storing one or more of the light input values as a calibration value when the feeder unit is absent from the feeder dock. When the feeder unit is absent from the feeder dock, the light beam can be transmitted from the light emitter to the light receiver substantially uninterrupted. Hence, the light input value that is measured during the calibration mode can be used as an indication of what the light receiver should measure as a light input value when the transparent section of the feeder unit is clean.

[0026] In another embodiment the control unit is configured for switching between the dynamic mode and a test mode, wherein the control unit, in the test mode, is configured for storing one or more of the light input values in a first situation in which the feeder unit is docked at the feeder dock and for storing one or more of the light input values in a second situation in which the feeder unit is removed from the feeder dock. The light input values from the different situations can be

compared, automatically or by the operator, to assess the condition of the feeder dock and the feeder unit. In particular, one can establish whether removing the feeder unit has a positive influence on the light input values, which could be an indication of the feeder unit being polluted. When the light input values in both situations are more or less the same, any decrease in the light input values during the dynamic mode can be contributed to pollution of the feeder dock.

[0027] In another embodiment the control unit is further configured for comparing the one or more light input values stored in the first situation with the one or more light input values stored in the second situation. Hence, the assessment discussed above can be made automatically.

[0028] In another embodiment the control unit is further configured for determining, based on the comparison, if one of the following conditions is true: [0029] a) one of the first situation and the second situation results in the light input values being different than in the other of the first situation and the second situation; or [0030] b) the light input values are the same or within a predetermined tolerance of each other for the first situation and the second situation. Again, the assessment discussed above can be made automatically.

[0031] In another embodiment the assembly comprises an alerting device for notifying an operator, wherein the control unit is operationally connected to the alerting device for sending a first notification to the operator via said alerting device when condition a) is true and for sending a second notification to the operator via said alerting device when condition b) is true. The operator can subsequently take appropriate action, for example cleaning one or both of the feeder dock and the feeder unit. When only the feeder unit is polluted, the feeder unit can be removed from the feeder dock without the need to inspect the feeder dock. Hence, the dispensing operation can be continued with minimal downtime, for example by docking a replacement feeder unit to the feeder dock while the removed feeder unit is being cleaned.

[0032] In another embodiment the control unit is configured for switching from the dynamic mode to the test mode when the light output of the light emitter is equal to or higher than an output threshold. Preferably, the output threshold is less than ninety percent of a maximum output of the light emitter. When said output threshold has been reached, it can be assumed that the sensor system, the feeder unit and/or the feeder dock have become too polluted to continue dispensing and detecting in a reliable manner. The test mode can then be triggered automatically.

[0033] It is noted that the test mode can also be executed independently from the dynamic mode, for example merely to do a check on the pollution of the feeder dock and/or the feeder unit, and can thus be made of divisional applications not requiring the control unit operating in a dynamic mode.

[0034] According to a third aspect, the invention provides a method for detecting events in an automatic dispensing process for dispensing discrete medicaments, wherein the method comprises the steps of: [0035] emitting light output from a light emitter of a detection pair; [0036] receiving light input by a light receiver of the detection pair and converting light input into one or more light input signals indicative of light input values; [0037] monitoring the light input values; and [0038] adjusting the light output of the light emitter in response to the light input values.

[0039] The method relates to the practical implementation of the sensor system according to the first aspect of the invention and/or the assembly according to the second aspect of the invention and thus has the same technical advantages, which will not be repeated hereafter.

[0040] In one embodiment the light output of the light emitter is increased when the light input values decrease over time.

[0041] In one embodiment the light output of the light emitter is increased when the light input values drop below an input threshold.

[0042] Preferably, the input threshold is a predetermined input threshold value.

[0043] Alternatively, the input threshold is a calibration value that is determined based on one or more of the light input values during a calibration mode.

[0044] In another embodiment the calibration value is determined when a light beam between the light emitter and the light receiver is unobstructed.

[0045] In another embodiment the input threshold is equal to or less than sixty percent of a maximum light input signal of the light receiver.

[0046] In another embodiment the light input values are monitored continuously or at a regular interval.

[0047] In another embodiment the light output of the light emitter is increased up to an output threshold.

[0048] In another embodiment the output threshold is less than ninety percent of a maximum output of the light emitter.

[0049] In another embodiment the method further comprises the step of: [0050] dispensing the discrete medicaments one-by-one at dispensing intervals, wherein the light input values are monitored for the one or more light input signals that are received between the dispensing intervals.

[0051] In another embodiment the method further comprises the steps of: [0052] providing a feeder dock for docking a feeder unit, wherein the light emitter and/or the light receiver are provided in, on or at the feeder dock; [0053] in a first situation docking the feeder unit at the feeder dock and monitoring one or more of the light input values when the feeder unit is docked at the feeder dock; [0054] in a second situation removing the feeder unit from the feeder dock and monitoring one or more of the light input values when the feeder unit is removed from the feeder dock; and [0055] comparing the one or more light input values monitored in the first situation with the one or more light input values monitored in the second situation.

[0056] It is again submitted that the method according to the embodiment described above can be applied independently of the step of increasing the light output when the light input values decrease over time, for example for the purpose of merely doing a check on the pollution of the feeder dock and/or the feeder unit.

[0057] In another embodiment the method further comprises the step of: [0058] determining, based on the comparison, if one of the following conditions is true: [0059] a) one of the first situation and the second situation results in the light input values being different than in the other of the first situation and the second situation; or [0060] b) the light input values are the same or within a predetermined tolerance of each other for the first situation and the second situation.

[0061] In another embodiment the method further comprises the steps of: [0062] sending a first notification to an operator when condition a) is true; and [0063] sending a second notification to an operator when condition b) is true.

[0064] In another embodiment the monitoring of the light input values in the first situation and the second situation and the comparison thereof are performed when the light output of the light emitter is equal to or higher than an output threshold.

[0065] Preferably, the output threshold is less than ninety percent of a maximum output of the light emitter.

[0066] According to a fourth aspect, the invention provides a computer program product comprising a non-transitory computer-readable medium holding instructions that, when executed by a processor, cause the sensor system according to the first aspect of the invention or the assembly according to the second aspect of the invention to perform the steps of the method according to the third aspect of the invention.

[0067] The computer program product can be used to implement the method according to the third aspect of the invention using the sensor system according to the first aspect of the invention and/or the assembly according to the second aspect of the invention and thus has the same technical advantages, which will not be repeated hereafter.

[0068] The various aspects and features described and shown in the specification can be applied, individually, wherever possible. These individual aspects, in particular the aspects and features described in the attached dependent claims, can be made subject of divisional patent applications.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0069] The invention will be elucidated on the basis of an exemplary embodiment shown in the attached schematic drawings, in which:

[0070] FIG. 1 shows a side view of an assembly of a sensor system for detecting events in an automatic dispensing process of a feeder unit and a feeder dock for docking said feeder unit; and

[0071] FIG. 2 shows side view of the assembly of FIG. 1 with the feeder unit received at said feeder dock;

[0072] FIG. 3 shows a flow chart of a method for operating the sensor system of FIGS. 1 and 2 in a dynamic mode; and

[0073] FIG. 4 shows a flow chart of a method for when switching between the dynamic mode of FIG. 3 and a test mode.

DETAILED DESCRIPTION OF THE INVENTION

[0074] FIGS. 1 and 2 shows an assembly 9 of a feeder dock 1 and a feeder unit 2 for use in an automatic dispensing process of or for dispensing discrete medicaments, in particular solid medicaments such as pills, tablets or capsules. The medicaments are 'discrete' or 'distinct' in the sense that they can be dispensed one-by-one or in dose-units.

[0075] The assembly 9 may be part of an automatic dispensing system, in particular the system according to U.S. Pat. No. 10,173,830 B1, having a plurality of feeder docks and feeder units, each with their own medicaments, to dispense and pack a wide variety of medicaments.

[0076] The feeder unit 2 may also be referred to as a 'medication dispensing container' or a 'canister'.

[0077] As shown in FIG. 1, the feeder dock 1 comprises a base 10 for receiving or docking the feeder unit 2, a dispensing drive 11 for engaging with and driving the feeder unit 2 and a plurality of positioning members 12, in particular positioning columns, for aligning and/or positioning the feeder unit 2 relative to the base 10 when docking. The feeder dock 1 further comprises an outlet channel 14. In the situation as shown in FIG. 1, the feeder unit 2 is removed from or not yet placed onto the feeder dock 1. FIG. 2 shows the situation when the feeder unit 2 has been docked to the feeder dock 1.

[0078] The feeder unit 2 comprises a supply chamber 21 for holding the discrete medicaments, and an outlet 24 for dispensing the discrete medicaments from the feeder unit 2 and a dispensing section 22 between the supply chamber 21 and the outlet 24 for feeding the discrete medicaments from the supply chamber 21 to the outlet 24. In particular, the dispensing section 22 comprises a dispensing mechanism 23, similar to the one disclosed in U.S. Pat. No. 10,173,830 B1, with one or more rotatable bodies with passageways for arranging the discrete medicaments in a single row, singulating the discrete medicaments and dispensing them one-by-one from respective waiting chambers into the outlet 24. When the feeder unit 2 is docked to the feeder dock 1, as shown in FIG. 2, the outlet 24 is aligned with and/or extends into the outlet channel 14 of the feeder dock 1.

[0079] The feeder unit 2 has an at least partially transparent housing 20. For example, the feeder unit 2 can have transparent sections of the housing 20 at the dispensing section 22 and/or the outlet 24.

[0080] The assembly 9 further comprises a sensor system 3 for detecting events in the automatic dispensing process. In this example, the sensor system 3 comprises a first light emitter 41 and a second light emitter 42 for generating a light output A. In particular, the first light emitter 41 and the second light emitter 42 generate a first light beam L1 and a second light beam L2, respectively. The light output A may be in the visible light spectrum or the invisible light spectrum. The light output A may for example be infrared or a laser light.

[0081] In this example, the light emitters 41, 42 are light emitting diodes (LED's). The light

emitters **41**, **42** may be part of a light curtain formed by several side-by-side light emitters, or the light emitters **41**, **42** may be in separate detection areas of the sensor system **3**, the feeder dock **1** and/or the feeder unit **2**.

[0082] The sensor system **3** further comprises a first light receiver **51** and a second light receiver **52**, for example in the form of photo cells. Each light emitter **41**, **42** forms a detection pair **41**, **51**; **42**, **52** with a respective one of the light receivers **51**, **52**. The light receivers **51**, **52** are arranged for receiving the light output A emitted by the respective light emitters **41**, **42** as a light input B and for converting said light input B into one or more input signals C, for example a single continuous input signal C that may vary in magnitude or a plurality of input signals C taken at certain intervals. The input signals are analogue, meaning that they represent a light input value D1, D2, . . . , Dn indicative of the strength and/or magnitude of the light input A that is being sensed, measured or detected. The light receivers **51**, **52** may be located opposite to the light emitters **41**, **42**, as shown in FIG. 1, at the same side as the light emitters **41**, **42**, in case reflectors (not shown) are being used.

[0083] The light emitters **41**, **42** and the light receivers **51**, **52** are arranged at, on or in the feeder dock **1**. They are positioned in such a way that the light beams L1, L2 travelling between the light emitter **41**, **42** and the light receiver **51**, **52** of the detection pairs **41**, **51**; **42**, **52** travel through a respective one of the transparent sections of the at least partially transparent housing **20** at least once. In this example, the first light emitter **41** and the first light receiver **51** have been positioned in the positioning members **12** on opposite sides of the dispensing section **22**. The second light emitter **42** and the second light receiver **52** have been positioned in the base **10** on opposite sides of the outlet **24**.

[0084] Alternatively, one or more of the light emitters **41**, **42** and/or the light receivers **51**, **52** may be positioned in the base **10**, the positioning members **12** or in dedicated protrusions extending along and/or at least partially into the feeder unit **2** from the feeder dock **1**. In a further alternative embodiment, the one or more of the light emitters **41**, **42** and/or the light receivers **51**, **52** may be arranged at, on or in the feeder unit **2**, either in a standalone manner or connected to the feeder dock **1** or another component when the feeder unit **2** is docked to said feeder dock **1**.

[0085] The sensor system **3** further comprises a memory **6** for storing the light input values D1, D2, . . . , Dn and a control unit **7** for processing the light input signals C or the light input values D1, D2, . . . , Dn, and for sending a light output value E to the light emitters **41**, **42** in response to the light input signals C or the light input values D1, D2, . . . , Dn in a manner that will be discussed hereafter in more detail. The memory **6**, in this example, is a computer-readable medium which is non-transitory or tangible, e.g. a physical data carrier such as a hard-drive, a USB-drive, a RAM memory or the like. A computer program product in the form of computer-readable instructions may be loaded onto the memory **6** and/or into the control unit **7**. The memory **6** can further be used to store the light input values D1, D2, Dn in a database like structure.

[0086] The control unit **7** is operationally connected to said one or more light emitters **41**, **42**, said one or more light receivers **51**, **52** and said memory **6**. The control unit **7** comprises a processor **8** for processing and/or carrying out computer readable instructions, for example instructions the memory **6** that cause the control unit **7** to carry out steps of a method according to the present invention.

[0087] The assembly **9** as a whole, or the sensor system **3** specifically, may further include an alerting device **90** for alerting an operator, for example a pharmacist, of a particular condition and/or actions to be taken. The alerting device **90** may be formed by a human machine interface, for example a display or a screen, or by one or more indicators, for example light and/or audio indicators. The alerting device **90** is used to indicate the state of the feeder dock **1** and/or the feeder unit **2**, for example whether the feeder dock **1** and/or the feeder unit **2** are clean or polluted.

[0088] A method for detecting events in the automatic dispensing process for dispensing discrete medicaments will now be briefly elucidated with reference to FIGS. 1-4.

[0089] FIG. 3 shows the steps of the method when the control unit 7 is operating in a dynamic mode M1.

[0090] At the start of the dynamic mode (step S1) a light input signal C or a light input value D1 is received from one of the light receivers 51, 52 (step S2) and processed by the processor 8 of the control unit 7. The control unit 7 is subsequently configured to check whether the light input value D1 has been received between dispensing intervals (Step S3). If the answer is 'yes' (Y) then the light input value D1 is stored in the memory 6 (step S4). If the answer is 'no' (N) then the method returns to the start (step S1).

[0091] The light input value D1 that is stored in the previous step (step S4) will hereafter be referred to as the 'last' light input value D1. Light input values D2, . . . , Dn stored during previous cycles of the method are referred to hereafter as 'previously stored' light input values D2, . . . , Dn.

[0092] In the next step of the method (step S5), the control unit 7 is instructed to compare the last light input value D1 with the previously stored light input value(s) D2, . . . , Dn. In particular, the control unit 7 checks (steps S6) whether the last light input value D1 has decreased compared to the previously stored light input value D2, . . . , Dn. When the answer is 'yes' (Y), the control unit 7 sends an instruction or a light output value E to the respective light emitter 41, 42 to increase the light output A (step S7). If the answer is 'no' (N), the last light input value D1 is stored in the memory 6 as one of the previously stored light input values D2, . . . , Dn and the steps S1 to S7 are repeated for the next 'last' light input value D1.

[0093] The control unit 7 may be configured to increase the light output value E or the light output A of the light emitter 41, 42 for any decrease in the last light input value D1 compared to the previously stored light input values D2, . . . , Dn. Alternatively, the control unit 7 may be configured for increasing the light output value E or the light output A of the light emitter 41, 42 only when the last light input values D1 drops below an input threshold. Said input threshold may be a predetermined input threshold value stored in the memory 6. It may for example be based on a percentage of the maximum light input signal of the light receiver 51, 52 or at the bottom end of an optimum working range of said the light receiver 51, 52. Such a value may for example be found in the factory specifications of the light receivers 51, 52. In this example, the input threshold is chosen to be equal to or less than sixty percent of a maximum light input signal of the light receiver 51, 52.

[0094] Alternatively, the light input threshold value may be set during a calibration mode, for example when the feeder unit 2 is removed and nothing is obstructing the light beam L1, L2. In this way, age degradation and/or small variations from factory specifications of the light emitter 41, 42 and/or the light receiver 51, 52 can be compensated for.

[0095] The light output A may also be decreased, for example when the light output A is higher than expected. This may occur after cleaning of the pollution or when the specifications of the respective light emitter 41, 42 are not in conformity with the factory specifications of said light emitter 41, 42. Hence, the light output A may be adjusted or corrected, both positively and negatively, to make sure that the respective light receiver 51, 52 can function in its optimal operating range or window.

[0096] The control unit 7, in the dynamic mode M1, is configured for increasing the light output A of the light emitter 41, 42 of the at least one detection pair 41, 51; 42, 52 up to an output threshold. The output threshold is less than ninety percent of a maximum output of the light emitter 41, 42 of the at least one detection pair 41, 51; 42, 52. The maximum output may be given as a factory specification or may be tested in the field.

[0097] FIG. 4 shows the steps of the method when the control unit 7 is operating in a test mode M2 or switched from the dynamic mode M1 to said test mode M2.

[0098] The steps of the method of FIG. 4 may overlap with or be executed in parallel to the steps of the method of FIG. 3. In particular, step S11 corresponds to step S1 which is the start of the dynamic mode M1. In one scenario, the next step (step S12) is also the same as step S2 of the previously discussed method in that the light input signal C or last light input value D1 is received

at or processed by the control unit 7. Alternatively, step S12 can be based on the light output value E of one of the light emitters 41, 42.

[0099] In the next step, it is determined whether the last light input value D1 is equal to or below an input threshold or whether the light output value E is equal to or above the aforementioned output threshold. If the answer is 'yes' (Y) the control unit 7 switches from the dynamic mode M1 to the test mode M2. If the answer is 'no' (N), the dynamic mode M1 continues as shown in FIG. 3.

[0100] At the start of the test mode M2 (step S14), the control unit 7 is configured for receiving a first light input value D1 in a first situation in which the feeder unit 2 is received at the feeder dock 1 (step S15), as shown in FIG. 2. If this is already the case, then no separate action is required. If this is not the case, the control unit 7 may send control signals to an automated member, for example a robot, to place the feeder unit 2 on the feeder dock 1, or it may alert the pharmacist to place the feeder unit 2 on the feeder dock 1. The method may involve additional checks (not shown) to check whether the feeder unit 2 has indeed been docked to or is present at the feeder dock 1. The first light input value D1 is stored in the memory 6.

[0101] Subsequently, the control unit 7 sends control signals to the automated member or instructions to the pharmacist to remove the feeder unit 2 from the feeder dock 1, to obtain the situation of FIG. 1 (step S16). The method may involve additional checks (not shown) to check whether the feeder unit 2 has indeed been removed or is absent from the feeder dock 1.

[0102] The control unit 7 is subsequently configured for receiving a second light input value D2 in a second situation in which the feeder unit 2 is removed from the feeder dock 1 (step S17). The second light input value D2 is also stored in the memory 6.

[0103] In the next step (step S18), the first light input values D1 representative of the first situation (FIG. 2), and the second light input value D2, representative of the second situation (FIG. 1), are compared to subsequently determine (step S19) whether the comparison result in a difference between the first light input value D1 and the second light input value D2. In particular, it is determined whether one of the following conditions is true: [0104] a) one of the first situation and the second situation results in the light input values D1, D2 being different than in the other of the first situation and the second situation; or [0105] b) the light input values D1, D2 are the same or within a predetermined tolerance of each other for the first situation and the second situation.

[0106] The predetermined tolerance may for example be less than ten percent of the first light input value D1 or less than five percent.

[0107] In particular, it is determined whether in condition a) the second light input value D2 is higher than the first light input value D1, which would be an indication that pollution of the feeder unit 2 is blocking or diffusing at least a part of the light beam L1, L2.

[0108] If condition a) is true, then the answer of step S19 is 'yes' (Y) and it is assumed that the change in light input value D1, D2 between the two situations is caused by the removal of the feeder unit 2. Consequently, the feeder unit 2 must be polluted. The control unit 7 is operationally connected to the alerting device 90 for sending a first notification N1 to the pharmacist via said alerting device 90. The first notification N1 may be indicative of a polluted state of the feeder unit 2.

[0109] If condition b) is true, then the answer of step S19 is 'no' (N) and it is assumed that the feeder dock 1 is polluted. The feeder unit 2 may also be polluted, but it is the feeder dock 1 that is so polluted that removing the feeder unit 2 does not result in a noticeable change in the light input values D1, D2. The control unit 7 sends a second notification N2 to the pharmacist via said alerting device 90. The second notification N2 may be indicative of a polluted state of the feeder dock 1.

[0110] It is to be understood that the above description is included to illustrate the operation of the preferred embodiments and is not meant to limit the scope of the invention. From the above discussion, many variations will be apparent to one skilled in the art that would yet be encompassed by the scope of the present invention.

TABLE-US-00001 LIST OF REFERENCE NUMERALS 1 feeder dock 10 base 11 dispensing

drive 12 positioning member 14 outlet channel 2 feeder unit 20 transparent housing 21 supply chamber 22 dispensing section 23 dispensing mechanism 24 outlet 3 sensor system 41 first light emitter 42 second light emitter 51 first light receiver 52 second light receiver 6 memory 7 control unit 8 processor 9 assembly 90 alerting device A light output B light input C light input signals D light input values E light output value L1 first light beam L2 second light beam M1 dynamic mode M2 test mode N1 first notification N2 second notification S1 step “start of dynamic mode” S2 step “receipt of light input signal” S3 step “has the light input signal been received between dispensing intervals?” S4 step “storing last light input value in the memory” S5 step “comparing last light input value with previously stored light input value(s)” S6 step “has the last light input value decreased compared to the previously stored light input value(s)?” S7 step “increase light output” S11 step “start of dynamic mode” S12 step “receipt of light output value or light input value” S13 step “is the light output value equal to or above an output threshold or is the light input value equal to or below an input threshold?” S14 step “start of test mode” S15 step “receiving the light input value in a first situation in which the feeder unit is received at the feeder dock” S16 step “removing the feeder unit from the feeder dock” S17 step “receiving the light input value in a second situation in which the feeder unit is removed from the feeder dock” S18 step “comparing light input values stored for the first situation and the second situation” S19 step “does the comparison result in a difference between the light input values stored for the first situation and the second situation?”

Claims

1. Assembly of a sensor system for detecting events in an automatic dispensing process for dispensing discrete medicaments and a feeder dock for docking a feeder unit, wherein the light emitter and/or the light receiver are provided in, on or at the feeder dock, wherein the feeder unit comprises a dispensing mechanism for dispensing the discrete medicaments one-by-one, wherein the feeder dock comprises a dispensing drive that is arranged for engaging with the dispensing mechanism when said feeder unit is received at the feeder dock, wherein the feeder unit has an at least partially transparent housing, wherein the light emitter and the light receiver are positioned such that, when the feeder unit is received in the feeder dock, a light beam travelling between the light emitter and the light receiver travels through a transparent section of the at least partially transparent housing at least once, wherein the sensor system comprises a light emitter for generating light output, a light receiver for converting light input into one or more light input signals indicative of light input values and a control unit that is operationally connected to said light emitter and said light receiver, wherein the light emitter forms a detection pair with the light receiver, wherein the control unit is configured for receiving the one or more light input signals from the light receiver and for operating in a dynamic mode in which the light output of the light emitter is adjusted in response to the light input values, wherein the light output of the light emitter increases when the light input values drop below an input threshold, and wherein the input threshold is a calibration value that is determined based on one or more of the light input values during a calibration mode, wherein the control unit is operationally connected to the dispensing drive to operate the dispensing mechanism at dispensing intervals, wherein the control unit, in the dynamic mode, is configured for storing the light input values for the one or more light input signals that are received between the dispensing intervals, wherein the control unit is configured for switching between the dynamic mode and a calibration mode, wherein the control unit, in the calibration mode, is configured for storing one or more of the light input values as a calibration value when the feeder unit is absent from the feeder dock, wherein the control unit is configured for switching between the dynamic mode and a test mode, wherein the control unit, in the test mode, is configured for storing one or more of the light input values in a first situation in which the feeder unit is docked at the feeder dock and for storing one or more of the light input values in a second situation in which the feeder unit is removed from the feeder dock, and wherein the control unit is

configured for switching from the dynamic mode to the test mode when the light output of the light emitter is equal to or higher than an output threshold.

2. Sensor system Assembly according to claim 1, wherein the control unit, in the dynamic mode, is configured for increasing the light output of the light emitter when the light input values decrease over time.

3. Sensor system Assembly according to claim 1, wherein the input threshold is a predetermined input threshold value stored in a memory and/or wherein the input threshold is equal to or less than sixty percent of a maximum light input signal of the light receiver.

4. (canceled)

5. Sensor system Assembly according to claim 1, wherein the control unit, in the dynamic mode, is configured: for storing the light input values continuously or at a regular interval; and/or for increasing the light output of the light emitter up to an output threshold.

6. (canceled)

7. Assembly according to claim 6, wherein the output threshold is less than or equal to ninety percent of a maximum output of the light emitter.

8-12. (canceled)

13. Assembly according to claim 1, wherein the control unit is further configured for comparing the one or more light input values stored in the first situation with the one or more light input values stored in the second situation.

14. Assembly according to claim 13, wherein the control unit is further configured for determining, based on the comparison, if one of the following conditions is true: a) one of the first situation and the second situation results in the light input values being different than in the other of the first situation and the second situation; or b) the light input values are the same or within a predetermined tolerance of each other for the first situation and the second situation.

15. Assembly according to claim 14, wherein the assembly comprises an alerting device for notifying an operator, wherein the control unit is operationally connected to the alerting device for sending a first notification to the operator via said alerting device when condition a) is true and for sending a second notification to the operator via said alerting device when condition b) is true.

16. (canceled)

17. (canceled)

18. Method for detecting events in an automatic dispensing process for dispensing discrete medicaments, wherein the method comprises the steps of: emitting light output from a light emitter of a detection pair; receiving light input by a light receiver of the detection pair and converting the light input into one or more light input signals indicative of light input values; monitoring the light input values over time; and adjusting the light output of the light emitter in response to the light input values, wherein the light output of the light emitter is increased when the light input values drop below an input threshold, and wherein the input threshold is a calibration value that is determined based on one or more of the light input values during a calibration mode.

19. Method according to claim 18, wherein the light output of the light emitter is increased when the light input values decrease over time.

20. Method according to claim 18, wherein the input threshold is a predetermined input threshold value.

21. Method according to claim 18, wherein the calibration value is determined when a light beam between the light emitter and the light receiver is unobstructed.

22. Method according to claim 18, wherein the input threshold is equal to or less than sixty percent of a maximum light input signal of the light receiver.

23. Method according to claim 18, wherein the light input values are monitored continuously or at a regular interval.

24. Method according to claim 18, wherein the light output of the light emitter is increased up to an output threshold.

- 25.** Method according to claim 24, wherein the output threshold is less than ninety percent of a maximum output of the light emitter.
- 26.** Method according to claim 18, wherein the method further comprises the step of: dispensing the discrete medicaments one-by-one at dispensing intervals, wherein the light input values are monitored for the one or more light input signals that are received between the dispensing intervals.
- 27.** Method according to claim 18, wherein the method further comprises the steps of: providing a feeder dock for docking a feeder unit, wherein the light emitter and/or the light receiver are provided in, on or at the feeder dock; in a first situation docking the feeder unit at the feeder dock and monitoring one or more of the light input values when the feeder unit is docked at the feeder dock; in a second situation removing the feeder unit from the feeder dock and monitoring one or more of the light input values when the feeder unit is removed from the feeder dock; and comparing the one or more light input values monitored in the first situation with the one or more light input values monitored in the second situation.
- 28.** Method according to claim 27, wherein the method further comprises the step of: determining, based on the comparison, if one of the following conditions is true: a) one of the first situation and the second situation results in the light input values being different than in the other of the first situation and the second situation; or b) the light input values are the same or within a predetermined tolerance of each other for the first situation and the second situation; wherein the method further comprises the steps of: sending a first notification to an operator when condition a) is true; and sending a second notification to the operator when condition b) is true.
- 29.** (canceled)
- 30.** Method according to claim 27, wherein the monitoring of the light input values in the first situation and the second situation and the comparison thereof are performed when the light output of the light emitter is equal to or higher than an output threshold.
- 31.** (canceled)
- 32.** Computer program product comprising a non-transitory computer-readable medium holding instructions that, when executed by a processor, cause the an assembly to perform the steps of the method according to claim 18.
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